

Duality for Representations of a Reductive Group over a Finite Field, II

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This paper is a continuation of [6] whose notations and references we shall preserve.

6

Let l be a prime number, $(l, q) = 1$, and let $\bar{\mathbb{Q}}_l$ be an algebraic closure of $\mathbb{Q}_l$. Let $\mathcal{R}(G)$ be the Grothendieck group of virtual G -modules over $\bar{\mathbb{Q}}_l$. We denote $D_G: \mathcal{R}(G) \rightarrow \mathcal{R}(G)$ the homomorphism such that

$$D_G(E) = \sum_i (-1)^i E_i,$$

for any finite-dimensional G -module E over $\bar{\mathbb{Q}}_l$.

Let T be a maximal torus of G , defined over $\mathbb{F}_q$ and let T be the group of its $\mathbb{F}_q$ -rational points. For any one dimensional T -module θ (over $\bar{\mathbb{Q}}_l$) we have defined in [5, 1.20] a virtual representation $R_T^G(\theta)$. Let $\sigma(G)$ be the $\mathbb{F}_q$ -rank of G .

We shall prove the following:

Theorem 0.1.

$$D_G(R_T^G(\theta)) = (-1)^{\sigma(G) - \sigma(T)} R_T^G(\theta).$$

Let us recall the definition of $R_T^G(\theta)$ or, more generally, that of “twisted induction” from Levi subgroups (cf. [5, 7]).

Let P be a parabolic subgroup of G , with unipotent radical U and let L be a Levi subgroup of P . We assume that L is defined over $\mathbb{F}_q$, but we make no assumption on P . Let L be the group of $\mathbb{F}_q$ -rational points of L . Let $F: G \rightarrow G$ be the Frobenius map corresponding to the $\mathbb{F}_q$ -rational structure of G . The variety

$$X_{L,P} = \{g \in G \mid g^{-1}F(g) \in P\}$$

has a $G \times L$ action

$$(g_0, l): g \mapsto g_0 g l^{-1}.$$

Hence we have a $G \times L$ -action $((g_0, l)^*)^{-1}$ of $G \times L$ on the cohomology groups $H_c^i(X_{L,P})$. (Here $H_c^i(\)$ denotes étale cohomology with compact support with coefficients in $\bar{\mathbb{Q}}_l$.) Let π be a finite-dimensional L -module over $\bar{\mathbb{Q}}_l$. Then $(H_c^i(X_{L,P}) \otimes \pi)_L$, with L acting diagonally, is in a natural way a G -module. By definition,

$$R_{L \subset P}^G(\pi) = \sum_i (-1)^i (H_c^i(X_{L,P}) \otimes \pi)_L \in \mathcal{R}(G).$$

In particular, if B is a Borel subgroup containing T , with unipotent radical V , the virtual representation $R_{T \subset B}^G(\theta)$ is defined. It is independent of B [5, 4.3] and is, by definition, $R_T^G(\theta)$.

Let $\langle \ , \ \rangle_G: \mathcal{R}(G) \times \mathcal{R}(G) \rightarrow \mathbb{Z}$ be the bilinear form defined by $\langle E, E' \rangle_G = \dim \text{Hom}_G(E, E')$ for any two finite-dimensional G -modules E, E' over $\bar{\mathbb{Q}}_l$.

The following result is a generalization of the orthogonality formula [5, 6.8].

Theorem 0.2.

$$\langle R_T^G(\theta), R_{L \subset P}^G(\pi) \rangle_G = \sum_{\substack{n \in G \\ n^{-1}Tn \subset L}} \langle R_{n^{-1}Tn \cap L}^L({}^n\theta), \pi \rangle_L,$$

where, for n in the sum, ${}^n\theta$ denotes the representation of $n^{-1}Tn$ defined by ${}^n\theta(n^{-1}tn) = \theta(t)$, $t \in T$.

7

We shall now give the proof of Theorem 6 assuming Theorem 7. If $G = T$, the theorem is obvious. Assume now that $G \neq T$ and that the theorem is already proved for groups of dimension $< \dim G$. We shall distinguish two cases.

Case 1. T is contained in a proper parabolic subgroup P of G , defined over $\mathbb{F}_q$. Let L be a Levi subgroup of P , containing T and defined over $\mathbb{F}_q$. Let L, P denote the groups of $\mathbb{F}_q$ -rational points of L, P . According to [5, 8.2] we have $R_T^G(\theta) = \text{Ind}(R_T^L(\theta))$, where $\text{Ind}: \mathcal{R}(L) \rightarrow \mathcal{R}(G)$ is the homomorphism defined by lifting a representation of L to P and then inducing it from P to G . According to [1], we have $\text{Ind} \circ D_L = D_G \circ \text{Ind}$. Since the theorem is assumed to be true for L , we have

$$\begin{aligned} D_G(R_T^G(\theta)) &= D_G \text{Ind}(R_T^L(\theta)) = \text{Ind}(D_L(R_T^L(\theta))) \\ &= \text{Ind}((-1)^{\sigma(L) - \sigma(T)} R_T^L(\theta)) = (-1)^{\sigma(G) - \sigma(T)} R_T^G(\theta), \end{aligned}$$

since $\sigma(G) = \sigma(L)$.

Case 2. For any proper parabolic subgroup P defined over $\mathbb{F}_q$, we have $T \not\subset P$.

Given such P , let L be a Levi subgroup of P defined over $\mathbb{F}_q$ and let P, L, Ind be defined as in Case 1. Our hypothesis shows that the set $\{n \in G \mid n^{-1}Tn \subset L\}$ is empty. By Theorem 7, we then have $\langle R_{L \subset P}^G(\pi), R_T^G(\theta) \rangle_G = 0$, for any L -module π of finite dimension. An argument entirely similar to that in the proof of [5, 8.2] shows that $R_{L \subset P}^G(\pi) = \text{Ind } \pi$. Thus we have

$$\langle \text{Ind } \pi, R_T^G(\theta) \rangle_G = 0$$

for any L -module π of finite dimension or, equivalently,

$$(R_T^G(\theta))^U = 0,$$

where $M \mapsto M^U$ is the homomorphism $\mathcal{R}(G) \rightarrow \mathcal{R}(L)$ such that, whenever M is a G -module, M^U is the space of vectors of M , invariant under U ($=$ the unipotent radical of P). From the definition of D_G , we see then that $D_G(R_T^G(\theta)) = (-1)^{\sigma(G)} R_T^G(\theta)$. It remains to use the fact that, for our T , we have $\sigma(G) - \sigma(T) \equiv 0 \pmod{2}$.

8

9

Let X be a scheme of finite type over an algebraic closure of $\mathbb{F}_q$ and let $f: X \rightarrow X$ be a morphism. We set

$$\mathcal{P}(f, X) = \sum_i (-1)^i \text{Tr}(f^*, H_c^i(X)).$$

In the proof of Theorem 7, we shall need the following:

Lemma 0.1. *Assume that f is an automorphism (of finite order) of X , and let T be a torus acting on X such that f commutes with each element of T . Then $\mathcal{P}(f, X) = \mathcal{P}(f, X^T)$, where X^T is the fixed point set of T on X .*

Proof. Write $f = f_u \cdot f_s = f_s \cdot f_u$, where f_s (resp. f_u) is a power of f and has order prime to p (resp. power of p). According to [5, 3.2], we have $\mathcal{P}(f, X) = \mathcal{P}(f_u, X^{f_s})$, $\mathcal{P}(f, X^T) = \mathcal{P}(f_u, (X^T)^{f_s}) = \mathcal{P}(f_u, (X^{f_s})^T)$. Thus, we are reduced to the case where f has order a power of p . We can find an element $t \in T$ such that $X^t = X^T$. Assume that we know that

$$\mathcal{P}(f, X) = \mathcal{P}(ft, X). \quad (1)$$

Using again [5, 3.2], we have $\mathcal{P}(ft, X) = \mathcal{P}(f, X^t) = \mathcal{P}(f, X^T)$; hence, by (1), we have $\mathcal{P}(f, X) = \mathcal{P}(f, X^T)$, as desired. It remains to prove (1).

We choose $\mathbb{F}_q$ -rational structures on T and X with Frobenius maps $F: T \rightarrow T$, $F: X \rightarrow X$, such that $F(t'x) = F(t')F(x)$, for all $t' \in T$, $x \in X$, $F(f) = f$ and $F(tx) = tF(x)$ for all $x \in X$. Using the trace formula for Frobenius maps, we see that (1) would be a consequence of

$$|X^{F^n f}| = |X^{F^n ft}| \quad \text{for all integers } n \geq 1. \quad (2)$$

Given $n \geq 1$, we can find $t_0 \in T$ such that $t_0^{-1} F^n(t_0) = t$. Then the map $x \mapsto t_0^{-1} x$ is a bijection $X^{F^n f} \xrightarrow{\sim} X^{F^n ft}$. This completes the proof of the lemma. $\square$

10

We shall now prove Theorem 7. As in the proof of [7, Theorem 8] we see that the left-hand side of (7.1) is equal to

$$|T|^{-1} |L|^{-1} \sum_{\substack{t \in T \\ l \in L \\ j \geq 0}} (-1)^j \text{Tr}((t, l), H_c^j(S)) \text{Tr}(t, \theta) \text{Tr}(l^{-1}, \pi), \quad (3)$$

where

$$S = \{(x, x', y) \in V \times U \times G \mid xF(y) = yx'\}$$

and $T \times L$ acts on S by

$$(t, l): (x, x', y) \mapsto (txt^{-1}, lx'l^{-1}, tyl^{-1}).$$

Let $C = \{n \in G \mid n^{-1}Tn \subset L\}$ and let $C_1, C_2, \dots, C_m$ be the orbits of L acting on C by right translation. For each i , $1 \leq i \leq m$, let $G_i = F^{-1}(V)C_iF^{-1}(P) = F^{-1}(V)C_iF^{-1}(U)$. Then G_i ($1 \leq i \leq m$) are the double cosets of G with respect to $F^{-1}(B)$ and $F^{-1}(P)$.

We set $S_i = \{(x, x', y) \in S \mid y \in G_i\}$. The S_i form a partition of S , stable under the action of $T \times L$. Corresponding to this partition, we have a decomposition of expression (3) into the sum of m terms, the terms being obtained from (3) by replacing S by S_i ($1 \leq i \leq m$). Let (3)(i) be the term corresponding to i . Let

$$\tilde{S}_i = \{(x, x', v, n, u) \in V \times U \times F^{-1}(V) \times C_i \times F^{-1}(U) \mid xF(vnu) = vlux'\}.$$

The map $\tilde{S}_i \rightarrow S_i$ defined by $(x, x', v, n, u) \mapsto (x, x', vnu)$ is a locally trivial fibration all of whose fibres are affine spaces of fixed dimension. Moreover, this map is $T \times L$ -equivariant, for the action of $T \times L$ on $\tilde{S}_i$ given by

$$(t, l): (x, x', v, n, u) \mapsto (txt^{-1}, lx'l^{-1}, tvn^{-1}, lnl^{-1}, lul^{-1}). \quad (4)$$

It follows that expression (3)(i) is equal to

$$|T|^{-1}|L|^{-1} \sum_{\substack{t \in T \\ l \in L \\ j \geq 0}} (-1)^j \text{Tr}((t, l), H_c^j(\tilde{S}_i)) \text{Tr}(t, \theta) \text{Tr}(l^{-1}, \pi). \quad (5)$$

We now make the change of variable $xF(v) \mapsto x$, $x'F(u)^{-1} \mapsto x'$. Then $\tilde{S}_i$ becomes

$$\{(x, x', v, n, u) \in V \times U \times F^{-1}(V) \times C_i \times F^{-1}(U) \mid xF(n) = vnx'\}.$$

The action of $T \times L$ is given by the same formula (4) as in the old coordinates. Next, we note that the same formula (4) defines an action of the group

$$H_i = \{(t, l) \in Z \times Z' \mid Z^{-1}F(l) = F(n_i)^{-1}t^{-1}F(t)F(n_i)t\}$$

on $\tilde{S}_i$, where Z (resp. Z') is the identity component of the centre of $n_iLn_i^{-1}$ (resp. of L) and n_i is any element of C_i . (Clearly, H_i does not depend on the choice of n_i in C_i .)

We shall now determine the fixed point set $(\tilde{S}_i)^{H_i^0}$ of the torus H_i^0 (= identity component of H_i) on $\tilde{S}_i$.

By Lang's theorem, the homomorphism $H_i \rightarrow Z$, $((t, l) \mapsto t)$ is onto, hence its restriction $H_i^0 \rightarrow Z$ must be also onto since Z is connected. Similarly, the homomorphism $H_i^0 \rightarrow Z'$, $((t, l) \mapsto l)$ is onto. Therefore, if $(x, x', v, n, u) \in (\tilde{S}_i)^{H_i^0}$, we must have

$$\begin{aligned} txt^{-1} &= x, & tvn^{-1} &= v & \text{for all } t \in Z, \\ lx'l^{-1} &= x', & lul^{-1} &= u & \text{for all } l \in Z'. \end{aligned}$$

Since L is the centralizer of Z' and $n_iLn_i^{-1}$ is the centralizer of Z , it follows that $x, v \in V \cap n_iLn_i^{-1}$ and $x', u \in U \cap L = \{1\}$. We then have $xF(n) = vn$, hence $F(n) = x^{-1}vn \in n_iLn_i^{-1}n$. Since $n \in C_i$, we have $F(n) \in n_iL$, hence $F(C_i) = C_i$. In particular, $(\tilde{S}_i)^{H_i^0}$ is empty, unless $F(C_i) = C_i$. Assume now that $F(C_i) = C_i$. Then $n_i \in C_i$ may be chosen so

that $n_i \in G^F$. The equation defining H_i may now be written: $F(ln_i^{-1}t^{-1}n_i) = ln_i^{-1}t^{-1}n_i$. It follows that

$$H_i^0 = \{(t, l) \in Z \times Z' \mid l = n_i^{-1}tn_i\}$$

and

$$(\tilde{S}_i)^{H_i^0} = \{(x, 1, v, n_i, 1) \mid x, v \in V \cap n_iLn_i^{-1}, y \in L, y^{-1}F(y) = n_i^{-1}v^{-1}xn_i\}.$$

Since the action of $T \times L$ on $\tilde{S}_i$ commutes with the action of the torus H_i^0 , expression (5) is equal, by Lemma 9, to

$$|T|^{-1}|L|^{-1} \sum_{\substack{t \in T \\ l \in L \\ j \geq 0}} (-1)^j \text{Tr}((t, l), H_c^j((\tilde{S}_i)^{H_i^0})) \text{Tr}(t, \theta) \text{Tr}(l^{-1}, \pi).$$

In the case where $F(C_i) = C_i$, the map $(\tilde{S}_i)^{H_i^0} \rightarrow \{y \in L \mid y^{-1}F(y) \in n_i^{-1}Vn_i\}$ given by $(x, 1, v, n_iy^{-1}, 1) \mapsto y$ is a locally trivial fibration all of whose fibres are affine spaces of the same dimension. The action of (t, l) on $(\tilde{S}_i)^{H_i^0}$ corresponds to the action $y \mapsto lyn_i^{-1}tn_i$. Hence expression (3)(i) is equal to

$$|T|^{-1}|L|^{-1} \sum_{\substack{t \in T \\ l \in L \\ j \geq 0 \\ F(C_i)=C_i}} (-1)^j \text{Tr}((t, l), H_c^j(\{y \in L \mid y^{-1}F(y) \in n_i^{-1}Vn_i\})) \text{Tr}(t, \theta) \text{Tr}(l^{-1}, \pi)$$

which is clearly equal to the right hand side of (7.1). This completes the proof of Theorem 6.

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